Searching in Databases

BioInf 2025/2026 1st Semester

André Lamúrias

a.lamurias@fct.unl.pt

Recap

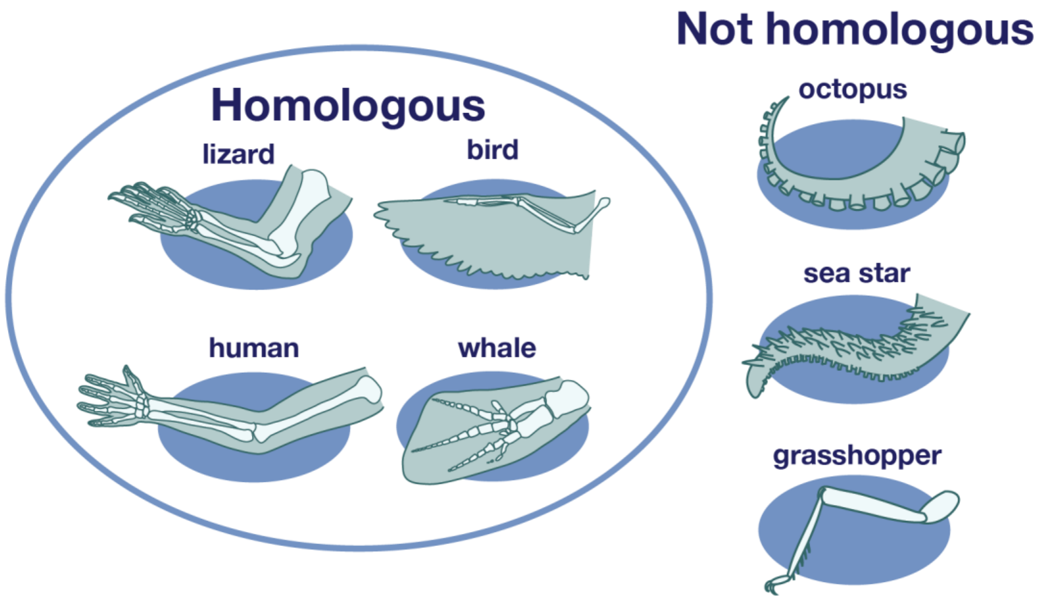
- Biological data is sequential (DNA, RNA, proteins)
- Biological processes transform DNA->RNA->proteins
- Computers can process these sequences and simulate the biological processes
- Computers can find patterns in biological sequences
- **Computers can align pairs of sequences to maximize an alignment score**

Searching Sequences in Databases

- Common request for a bioinformatician: provide some **annotations** to a new sequence
  - Annotations: descriptions of what it does
- Trying to infer the function from similar sequences is a way to address this task:
  - identify sequences that are similar to the target (query) sequence which are already annotated with a given function
- **Scan databases** containing annotated sequences and compare our query sequence against all sequences in the database, seeking to find the ones with **higher similarity**

Searching Sequences in Databases

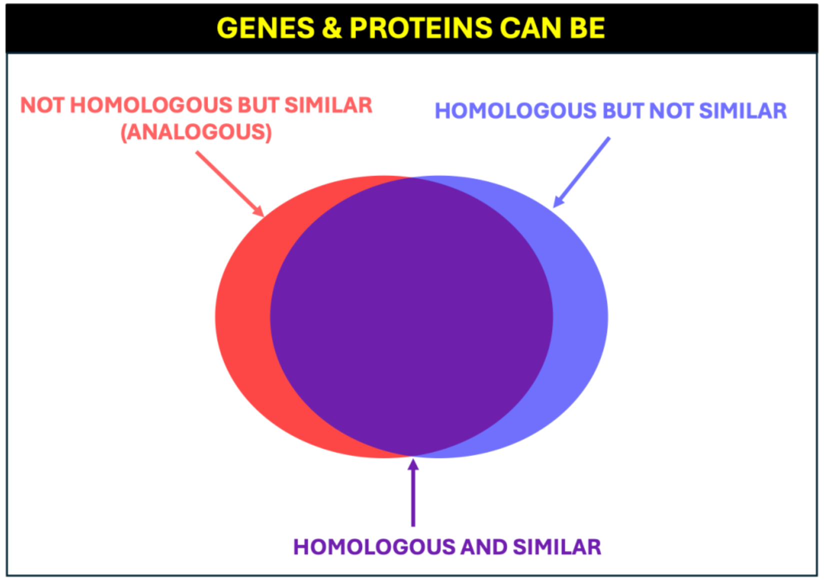
- Major bioinformatics objective: elucidate the function of given genes or the proteins they encode.
- Generate hypotheses regarding biological function, based on **similarity** of sequences to others that have a determined function
  - preferably experimentally validated.
- We want to infer **homology**, i.e. the existence of shared ancestry, based on sequence similarity.

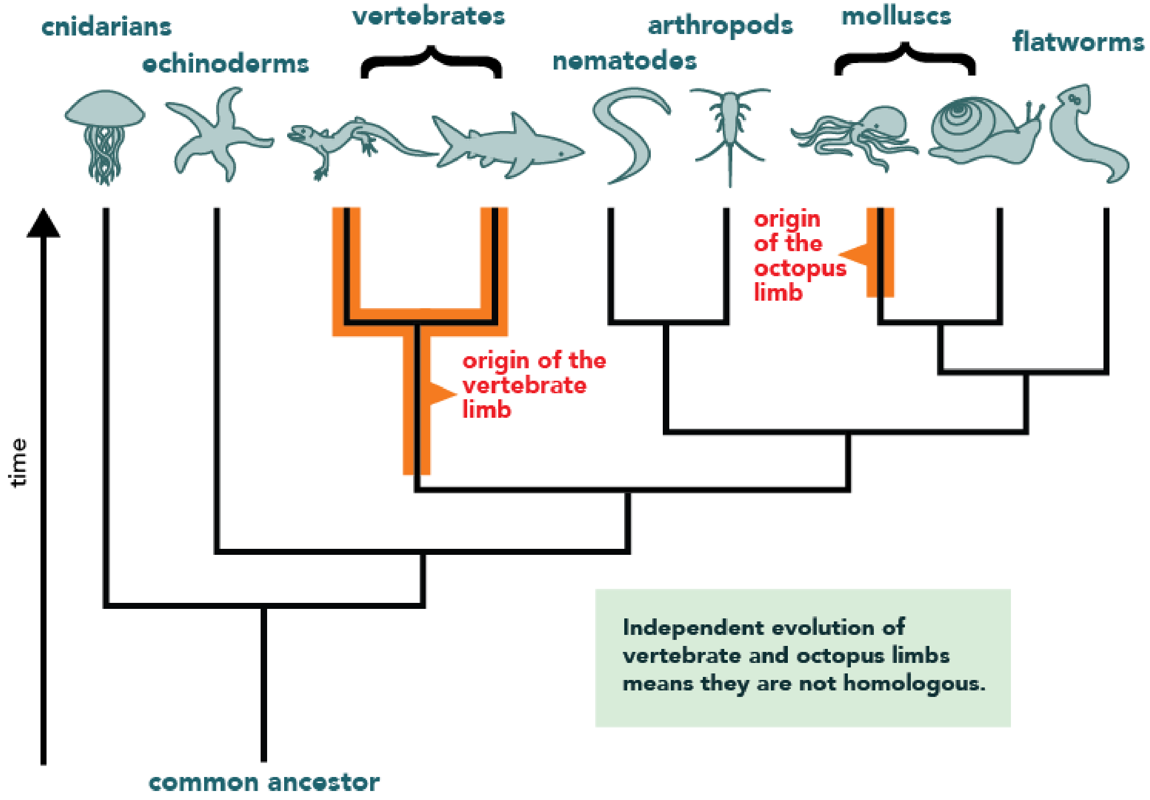


https://evolution.berkeley.edu/similarities-and-differences-understanding-homology-and-analogy-ms/not-all-similarity-is-homology/

Searching Sequences in Databases

- Similarity and homology are not equivalent
  - Similarity – continuous variable; Homology - discrete
  - but sequences that show a high degree of similarity have a high probability of being homologous and sharing similar functions.





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Searching Sequences in Databases

- If we have a database with sequences for which we know their function:
  - Run the pairwise sequence alignment algorithms to compare the query with each of the sequences in the database and select the one(s) with the highest score.

- It is not an adequate approach!!

Searching Sequences in Databases

- Searching databases requires many comparisons
  - GenBank has 230 million and WGS 1.6 billion sequences
  - Even though dynamic programming alignment is fast, we need much faster algorithms
- Finding matches requires statistics:
  - when **aligning** two sequences we assume they are **evolutionarily related** and measure their relatedness with the alignment score.
  - when we **search** a sequence database, we want to decide whether or not any sequence in the database is **significantly similar** to the query sequence.
- We are interested in finding sequences with **conserved regions** and not necessarily sequences that are all uniformly similar to the query sequence.

Searching Sequences in Databases

- Alignment algorithms such as Smith-Waterman are too slow for the query sequence lengths and databases sizes
- Basic heuristic: find short exact matches and use only those
  - Can also allow for some mismatches
  - e.g. filter out database sequences that do not share at least one 4-mer with the query
- Generalizes Approximate Pattern Matching
  - But we do not need to match the whole query
  - We want to match substrings of the query

Searching Sequences in Databases

- The objective of a sequence search algorithm is not the same as a sequence alignment:
    - Sequence alignment presupposes that the sequences are homologous
    - Sequence search finds sequences that are likely to be homologous to the query sequence.
- Although the sequence comparison algorithms used to search databases also perform alignments, they are not optimal alignment algorithms:
    - they are not intended to find the best alignment
    - they are meant to estimate homology through sequence similarity.
- Do not use the results of sequence searches as if they were optimal alignments
    - for a good alignment realign with the appropriate algorithm

Sequence Search Algorithms

Query Matching Problem:
Find all substrings of the query that approximately match the text.

Input: Query $\mathbf{q} = q_1 \dots q_p$, text $\mathbf{t} = t_1 \dots t_m$, and integers n and k.

Output: All pairs of positions (i, j) where $1 \le i \le p - n + 1$ and $1 \le j \le m - n + 1$ such that the n-letter substring of $\mathbf{q}$ starting at i approximately matches the n-letter substring of $\mathbf{t}$ starting at j, with at most k mismatches.

Sequence Search Algorithms

- A few heuristic algorithms have been proposed by the Bioinformatics community over the last years, to address sequence search in large databases.
- The most popular ones are:
  - FASTA, that appeared in the early days of Bioinformatics, resulting in a well known file format
  - BLAST (Basic Local Alignment Search Tool) that is now a widely used program available to be run by the community in a large number of servers.

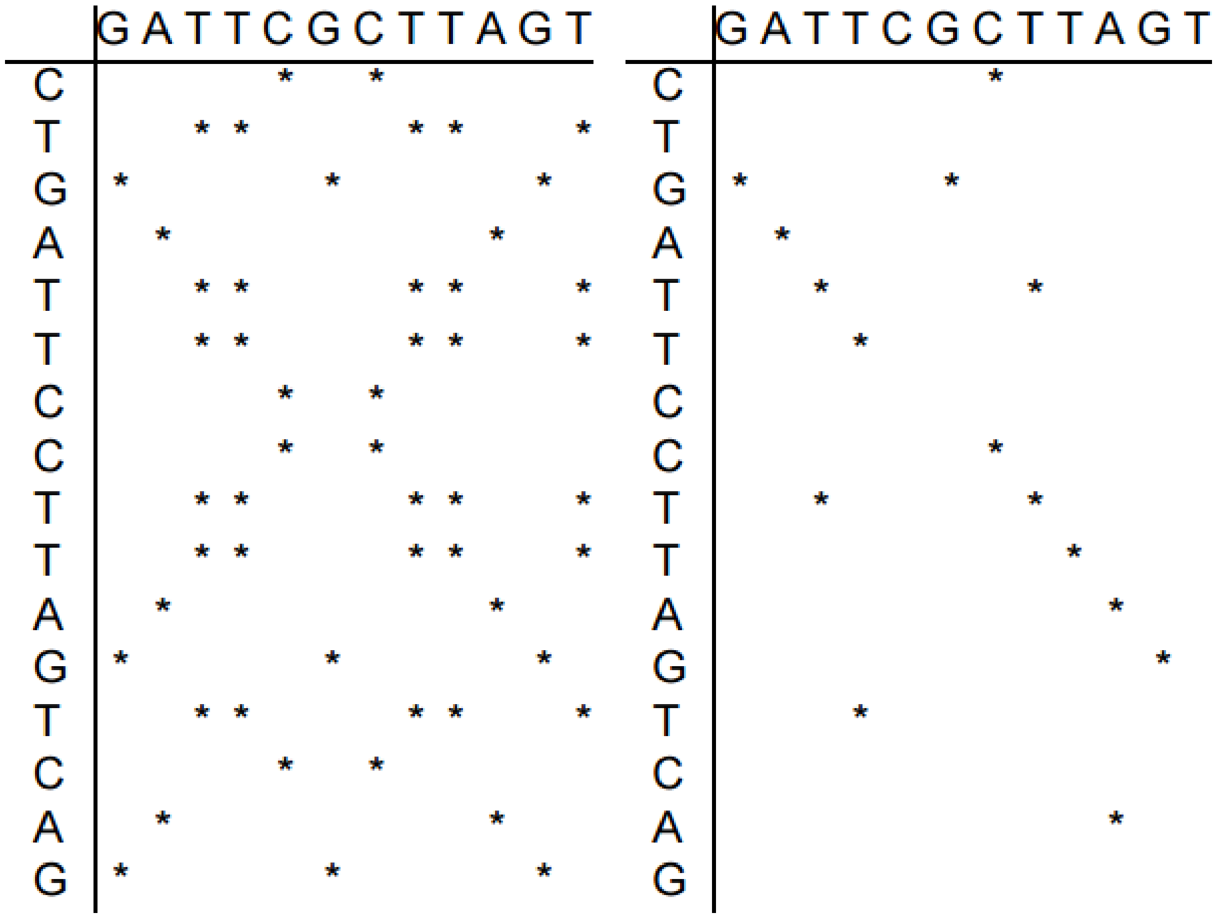
Sequence Search Algorithms

FASTA

- FASTA is a fast algorithm for finding similar sequences
  - It is not an optimal alignment algorithm.
  - The heuristics make the computation faster but do not guarantee optimality
- Focuses on the best matching stretches of the sequence
  - then builds an alignment that follows these stretches within a limited window, thus reducing the computational demands.

FASTA

- Select diagonal with high concentration of matches
- In the second dot-plot, we put a dot if two positions match: GA in the vertical sequence matches GA in the horizontal sequences
- Smoothing



Sequence Search Algorithms

BLAST (Basic Local Alignment Search Tool)

- currently the most used to perform searches over DNA or protein sequence databases.
- similar to FASTA, but around 100 times faster
    - It gains this advantage by skipping the alignment step and focusing on high-scoring pairs HSPs (also known as maximal segment pairs MSPs)

BLAST

- Whose genome is the following sequence encoded in? What is its function?
- Use blastp: https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastp&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

```
MFVFLVLLPLVSSQCVNLTTRTQLPPAYTNSFTRGVYYPDKVFRSSVLHSTQDLFLPFFS
NVTWFHAIHVSGTNGTKRFDNPVLPFNDGVYFASTEKSNIIRGWIFGTTLDSKTQSLLIV
NNATNVVIKVCEFQFCNDPFLGVYYHKNNKSWMESEFRVYSSANNCTFEYVSQPFLMDLE
GKQGNFKNLREFVFKNIDGYFKIYSKHTPINLVRDLPQGFSALEPLVDLPIGINITRFQT
LLALHRSYLTPGDSSSGWTAGAAAYYVGYLQPRTFLLKYNENGTITDAVDCALDPLSETK
CTLKSFTVEKGIYQTSNFRVQPTESIVRFPNITNLCPFGEVFNATRFASVYAWNRKRISN
CVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTNVYADSFVIRGDEVRQIAPGQTGKIAD
YNYKLPDDFTGCVIAWNSNNLDSKVGGNYNYLYRLFRKSNLKPFERDISTEIYQAGSTPC
NGVEGFNCYFPLQSYGFQPTNGVGYQPYRVVVLSFELLHAPATVCGPKKSTNLVKNKCVN
FNFNGLTGTGVLTESNKKFLPFQQFGRDIADTTDAVRDPQTLEILDITPCSFGGVSVITP
GTNTSNQVAVLYQDVNCTEVPVAIHADQLTPTWRVYSTGSNVFQTRAGCLIGAEHVNNSY
ECDIPIGAGICASYQTQTNSPRRARSVASQSIIAYTMSLGAENSVAYSNNSIAIPTNFTI
SVTTEILPVSMTKTSVDCTMYICGDSTECSNLLLQYGSFCTQLNRALTGIAVEQDKNTQE
VFAQVKQIYKTPPIKDFGGFNFSQILPDPSKPSKRSFIEDLLFNKVTLADAGFIKQYGDC
LGDIAARDLICAQKFNGLTVLPPLLTDEMIAQYTSALLAGTITSGWTFGAGAALQIPFAM
QMAYRFNGIGVTQNVLYENQKLIANQFNSAIGKIQDSLSSTASALGKLQDVVNQNAQALN
TLVKQLSSNFGAISSVLNDILSRLDKVEAEVQIDRLITGRLQSLQTYVTQQLIRAAEIRA
SANLAATKMSECVLGQSKRVDFCGKGYHLMSFPQSAPHGVVFLHVTYVPAQEKNFTTAPA
ICHDGKAHFPREGVFVSNGTHWFVTQRNFYEPQIITTDNTFVSGNCDVVIGIVNNTVYDP
LQPELDSFKEELDKYFKNHTSPDVDLGDISGINASVVNIQKEIDRLNEVAKNLNESLIDL
QELGKYEQYIKWPWYIWLGFIAGLIAIVMVTIMLCCMTSCCSCLKGCCSCGSCCKFDEDD SEPVLKGVKLHYT
```

BLAST

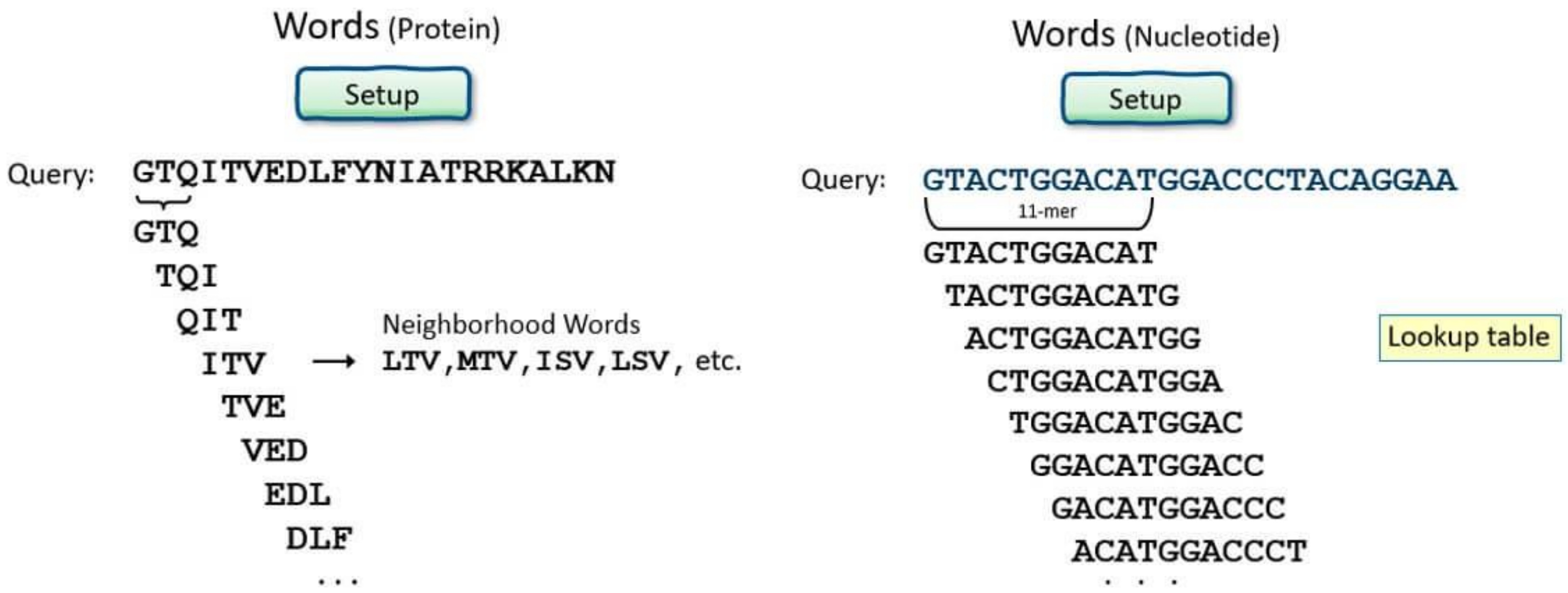
- Use short “words” (i.e. sub-sequences) and search for matches of high similarity (with no gaps) of these words in the **query** and in the **sequences from the database**.
- Extended, in both directions, to obtain high-quality alignments of larger dimension.
- Use statistics for estimate significance of found matches

BLAST

- The main steps of the BLAST algorithm are:

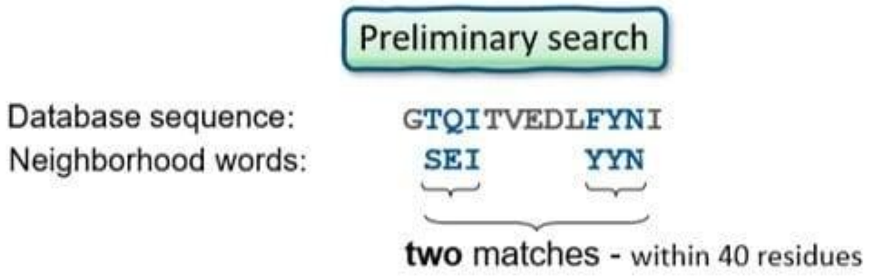
BLAST

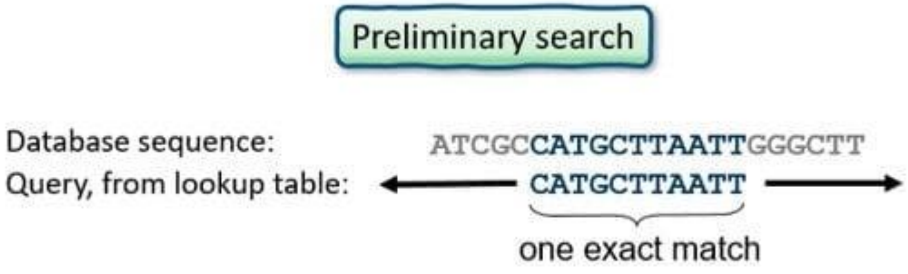
- Step 1: Remove **regions of low complexity**
- Step 2 and 3: create list of words and compile similar words



BLAST

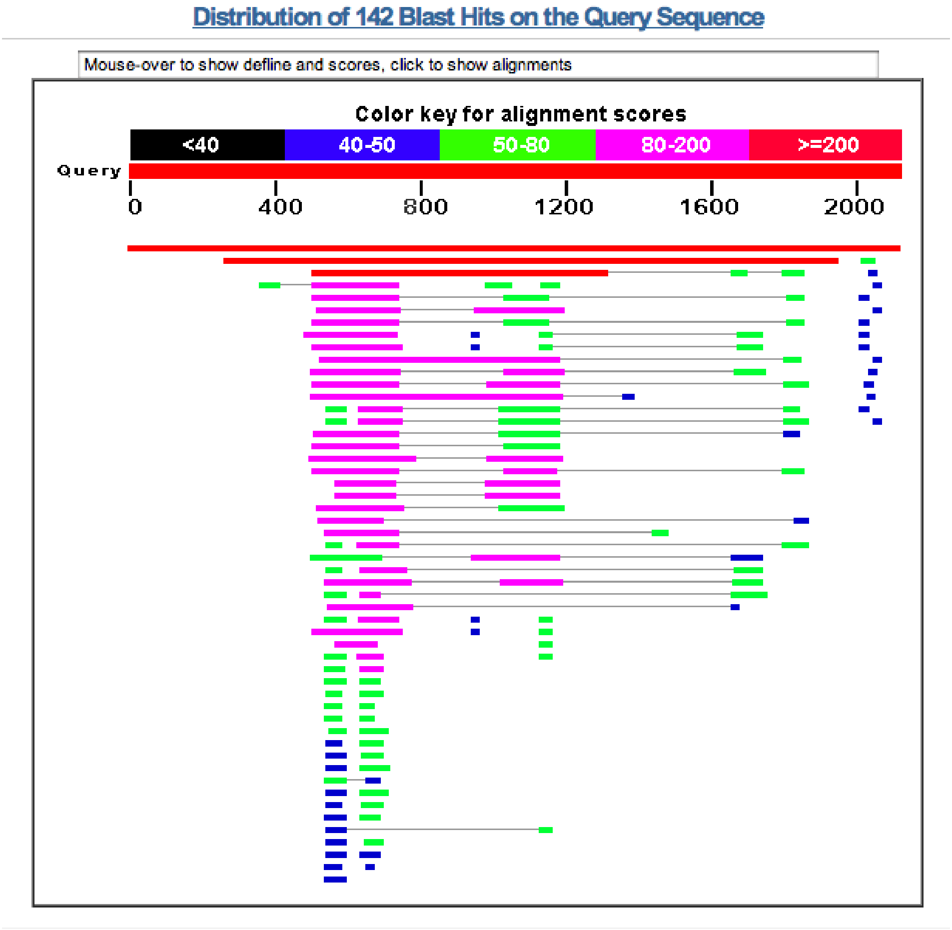
- Step 4: search database for sequences with those words





BLAST

- Step 5 and 6: Expand and select HSPs



BLAST

- The main steps of the BLAST algorithm are:
  1. Remove **regions of low complexity** (e.g. sequence repeats) from the sequence (may compromise the quality of the alignment)
  2. Obtain **all possible “words”** of size w, i.e. sub-sequences of length w occurring in the **query sequence**;
  3. For each word from the previous step, compile the list of **all possible words of size w** that can be defined in the allowable alphabet, whose **alignment score** (no gaps, using SM) **is higher than a threshold** T;
  4. Search in the database for **all occurrences of the words** collected in the last step, which represent matches (**hits**) of size w between the query and one of the database sequences;
  5. **Extend all hits** from the last step, in both directions, while the score follows a given criterion (typically, the criterion is dependent on the size of the extension);
  6. **Select the alignments** in the previous step with highest scores, normalized for its size (these are named the **high-scoring pairs-HSPs**).

BLAST programs

- Programs specific for different sequence types
- can be locally installed or run over servers in the web: most important is the NCBI (National Center for Biotechnology Information) at:
  - https://www.ncbi.nlm.nih.gov/BLAST
- `blastn` searching nucleotide sequences in nucleotide databases
- `blastp` searching protein in protein databases
- Both include a set of adjustable parameters with different default values:
  - word size `w` `(blastn 11; blastp 6)`
  - scoring function `(blastn match/mismatch:2/-3; blastp BLOSUM62)`
  - gap penalties `(blastn g/r:-5/-2; blastp g/r:-11/-1)`

BLAST programs

- Other programs in the BLAST suite which may be used to search for sequences of a different type:
  - `blastx`: takes a **DNA sequence as query**, but **searches over protein sequences**, and thus can be used to find potential protein products encoded by a nucleotide sequence;
  - `tblastn`: takes a **protein sequence** as query, but **searches over DNA sequence** databases, thus trying to identify database sequences encoding proteins similar to the one in the query;
  - `tblastx`: takes a **DNA sequence as input**, searching over DNA sequence databases, but in both cases the **sequences are translated** and the **matches are searched over protein sequences**; this method can be used to identify nucleotide sequences similar to the query based on their coding potential.

BLAST through Biopython

- Can run locally or through remote servers
  - Most commonly: NCBI servers

```
from Bio.Blast import NCBIWWW
from Bio import SeqIO
record = SeqIO.read(open("example_blast.fasta"), format="fasta")
result_handle = NCBIWWW.qblast("blastn", "nt", record. format("
fasta"))

save_file = open("my_blast.xml", "w")
save_file.write(result_handle.read())
save_file.close()
result_handle.close()

result_handle = open("my_blast.xml")
```

BLAST through Biopython

- Can run locally or through remote servers
  - Most commonly: NCBI servers

```
from Bio.Blast import NCBIXML
blast_record = NCBIXML.read(result_handle)
blast_records = NCBIXML.parse(result_handle)
E_threshold = 0.001
for blast_record i n blast_records:
    for alignment i n blast_record.alignments:
       for hsp i n alignment.hsps:
          if hsp.expect < E_threshold:
             print ("****Alignment****")
             print ("sequence:", alignment.title)
             print ("length:", alignment.length)
             print ("e value:", hsp.expect)
             print (hsp.query[0:75] + "..." )
             print ( hsp.match[0:75] + "...")
             print ( hsp.sbjct[0:75] + "...")
```

Statistical Significance of the Alignments

- Scoring functions used by BLAST and dynamic programming, based on substitution matrices and gap penalties, are **relative to the size of the sequences**.
- Useful to compare different alignments for the same pair of sequences, but cannot be used to compare alignments of different sequences with distinct sizes.
- Comparing hits of a query with different sequences from a database, requires to compute normalized scores.

Statistical Significance of the Alignments

- To be able to infer possible homology from sequence similarity we need to answer the following questions:
  - is the similarity found between the query and the sequence, for a given alignment, statistically significant?
  - how probable is that this similarity occurs by pure chance?
- The E-value is a metric to evaluate the significance of the alignment, provided as a result for each HSP in BLAST
- $E = kmNe^{-\lambda s}$ where k and λ are constants, m is the number of letters in the query, N is the total number of letters in the target database, and s is the score of the high-scoring segment pair.

Statistical Significance of the Alignments

- Indicates the number of expected alignments with a score at least as high as the one provided by the current HSP.
- The lower the value of E is, i.e. the closer it is to zero, the more significant the considered match is.
- Results are filtered by a maximum allowed E-value
  - values from 10^{-5} up to 0.05 used in different scenarios and by different authors.

Summary

- Searching Sequence Databases
- Sequence Search Algorithms
    - FASTA
    - BLAST
- Using BLAST
- Statistical Significance of the Alignments

Further reading

- Rocha Chapter 7
- Jones Chapter 9
- Baxevanis Chapter 3